



Bikes that don't crash

Ducati has developed a system that ensures their bikes don't crash as much as possible. Read more: <http://dgit.in/dcaticrsh>



The apple is for everyone

Apple is making Swift, their programming language, open source for iOS, OSX and Linux. <http://dgit.in/applopen>

From the labs



More juice please

With portable battery-powered devices still ruling our lives, it's about time we cracked the battery code. Will we ever measure battery life in weeks and years instead of hours?

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Since the 18th century, when man first discovered how to harness static electricity using metal electrodes and saltwater, the quest for improving batteries has become an important field of study for many. From those days of yore, battery tech has undergone significant improvements, however, the basic chemistry remains the same. A basic battery today contains three major parts: the anode, cathode and separator (also known as “electrolyte”). The anode and cathode provide the electrons that conduct electricity, while the separator

facilitates the reaction between the metals at both ends of the battery, i.e. the anode and cathode. In the search for better batteries, the only thing that has changed is the material used for the cathode, anode and separator.

Science and Technology, on the larger scale, has made tremendous strides. We have laptops, smartphones and tablets that are many orders of magnitude more powerful than the first computers invented. And the best part: they're not devices that are locked to a wall plug. Thanks to batteries, they're portable and we've come to depend on them. Imagine the world today with landlines instead of mobile phones!

Over time, more device categories went portable. However, along with becoming more powerful, they've also gotten more power-hungry. Batteries meanwhile are still playing catch up. Better performing batteries with longer life continue to evade us as we rush to find the nearest plug point to recharge this much needed life force at the most inopportune times.

In search of the ultimate battery

Meanwhile, scientists seem to have gone through nearly all elements on our periodic table to find more viable and efficient metals for use in batteries. Is the battery that we use today in our laptops and mobiles the ultimate battery?

These scientists are now trying to improve on the widely used battery of today – the lithium-ion battery. This is one line of work. Yet another group of researchers is trying to develop rechargeable batteries that will harness the power of nature by way of sound, light and even our own muscles. Others are trying to make devices more power efficient.

Whatever be the method and approach, all these endeavors converge into one ultimate goal – developing methods to let us work seamlessly without having to worry about running out of battery.

Batteries of the future

The lithium-ion batteries that we use today feature lithium cobalt oxide as the cathode and carbon as the anode, separated by a very thin sheet of microperforated plastic. Lightweight and with high power density, a Li-ion cell produces a voltage of 2.7 volts, which is much higher than a normal AA alkaline cell, making it right choice for use in mobile devices. However, there are a few downsides to Li-ion batteries. They begin degrading as soon as they leave their manufacturing factories, are extremely sensitive to high temperatures and might burst into flames on failure.

Thus, arises the need for batteries that can overcome these disadvantages while increasing power density. Let's take a look at some of the research underway to improve battery tech.